Z-F-002164(Army-GL)
March 14, 1983

INTERIM FEDERAL SPECIFICATION

FRUIT COCKTAIL, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers thermostabilized tray pack fruit cocktail in heavy syrup for use by the Department of Defense as a component of operational rations.

1.2 Classification. The tray pack fruit shall be of the following types as specified (see 6.1).

Type I - Processed from uncanned fruit cocktail.
Type II - Processed from canned fruit cocktail.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

FSC 8915

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-L-35078 - Loads, Unit, Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products

U.S. Standards for Grades of Canned Fruit Cocktail

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.4).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors and off-colors.

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3.2.1 Fruit. For type I product, the fruit shall be fresh, mature seedless grapes, peaches, pears, pineapple, and maraschino cherries. For type II product the canned fruit cocktail shall be U.S. Grade A fruit cocktail packed in heavy syrup (18 to 22° Brix) of the latest season's pack.

3.2.2 Sweetening ingredients. The sweetening ingredients shall be a blend of corn syrup, white, refined, granulated cane or beet sugar, or liquid sugar, or any combination of invert sugar or dextrose to produce a 75-80 percent corn syrup - 20-25 percent sugar blend. The sweetening ingredients shall conform to MIL-STD-900.

3.2.3 Citric acid. Citric acid shall be in accordance with the requirements of the Food Chemicals Codex.

3.3 Fruit processing. Seedless grapes shall be thoroughly washed and drained. Peaches, pears, and pineapple shall be thoroughly washed, peeled, and diced in 10-mm cubes, and blanched. Maraschino cherries shall be artifically colored, halved, rinsed and drained.

3.4 Syrup processing. Sweetening ingredients, citric acid and water shall be the only ingredients used in preparation of the heavy syrup.

3.5 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the following requirements:

a. The prepared fruit and syrup shall be mixed in the following approximate proportions:

Ingredient	Percent by weight
Peaches	30
Pears	25
Grapes	10
Pineapple	7
Cherries	3
Syrup	25
	100

b. The average net weight shall be 106 ounces minimum.
c. No individual can shall contain less than 104 ounces of product.
d. The average drained weight of fruit shall be 70 ounces minimum.
e. No individual can shall contain less than 68 ounces of drained fruit.
f. The can shall be hermetically sealed immediately after filling.

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3.6 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until the temperature at the center of the product reaches 195°F. Records shall be maintained for each lot.

3.7 Finished product requirements. The finished product shall meet the requirements for canned fruit cocktail, in heavy syrup (18° to 22° Brix), and shall be U.S. Grade A of the U.S. Standards for Grades of Canned Fruit Cocktail. The pH of the finished product shall be a maximum of 4.3.

3.7.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.8 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of applicable government inspection agency.

3.9 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance to the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade Certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, and sealing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The Government reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by the AMS agent.)

4.6.3 pH testing. Three tray packs shall be selected at random from the lot. The contents of each tray pack shall be blended separately in a Waring blender or equivalent to form the sample. The pH of the sample shall be determined in accordance with the Official Methods of Analysis of the Association of Analytical Chemists. If the pH of the sample is more than 4.3, the lot shall be rejected.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers.

4.8 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

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Major: National stock number, item description, contract number or date of pack markings missing, incorrect or illegible. Minor: (1) Other required markings missing, incorrect or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used. Level C shipping containers specified in 5.2.3 shall be examined only for the marking defects heretofore specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product of one type, sealed, and thermoprocessed as specified in 3.5 and 3.6. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, of one type, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, of one type, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

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5.2.3 Level C packing. Four cans of product, of one type, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2.4 Alternate level C packing. When specified (see 6.1), four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension and be fabricated of class domestic fiberboard.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to Navy and Air Force, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each package shall be labeled in accordance with MIL-L-1497. In addition each package shall specify type of product. Paper labels are not permitted. Cans shall show the following statements:

DO NOT HEAT

YIELD

Approximately 25 servings of 4 each

5.5. Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of product required (see 1.2).
(c) When a first article is required (see 3.1, 4.3, and 6.4).

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(d) Provisions for approved preproduction samples (see 3.7.1 and 6.4).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(g) When alternate (special requirements) level C packing is needed (see 5.2.4).
(h) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article shall be a preproduction sample. The contracting officer should include specific instructions in all procurement instruments regarding arrangement for selection, inspection and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A656

☆ U.S. GOVERNMENT PRINTING OFFICE: 1983—605-034/1834

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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